

Gas Absorption in a Packed Bed

Pre-Lab Plan

Unit Operations Lab #1

Date: January 21st, 2020

Group #2 Tuesday 9 AM section

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Erik Yspango

Commented [PTH1]: 29.5/35

1. Experimental Objective

There are two objectives in this experiment. One, is to determine the fluid flow characteristics in a packed bed gas absorber as well as the flooding point. Two, is to determine gas absorption characteristics and the overall mass transfer coefficient in the removal of CO₂ from an air stream with water in a packed bed gas absorber.

Commented [PTH2]: 3.5/5

Commented [PTH3]: Missing info here. There are three parts to this experiment and you should have summarize all of them in the objective.

Commented [PTH4]: 4/5

2. Experimental

2.1 Apparatus

The packed absorption column is made up of two diameter clear sections which are connected and installed vertically. The column is filled with glass Raschig rings, but many types of packing can be used in gas absorption. Liquid for the process is stored in a large rectangular feed tank at the bottom and a pump is used to deliver the liquid to the top of the column where it falls through the packing then returns back to the tank.

Commented [AZ5]: Good

Commented [PTH6]: ok but figure caption should be below the figure.

Figure 1: Gas absorption column-UOP7

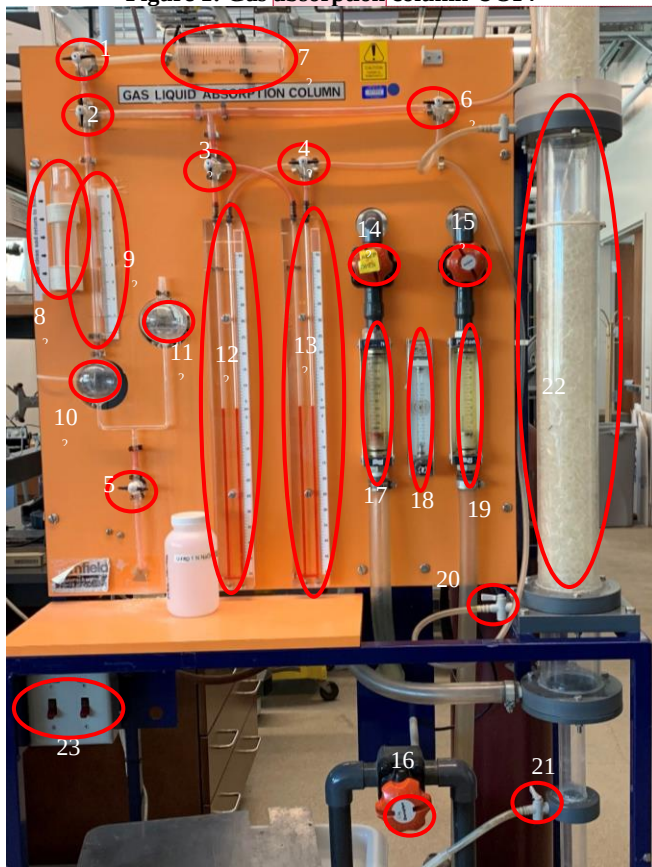


Figure 2: Gas absorption column backside-UOP7



Table 2.1: Apparatus Descriptions

No.	Name	Description
1-6	Three-Way Valves	Stops flow from one tube, and pushes it into another
7	Volume Measuring Tube	Part of the Hempel apparatus. Measure the amount of air in the tube
8	Plunger	Part of the Hempel apparatus. Used to push the air into the bulbs.
9	NaOH Measurement	Part of the Hempel apparatus. Measure the amount of CO ₂ in the air
10-11	Bulbs	Part of the Hempel apparatus. Holds the NaOH at initial concentration.
12-13	Manometer	Measures the pressure differential between top and middle or top and bottom
14	“Do Not Touch” Valve	Turns the flowrate of the Carbon Dioxide and Water off/on respectively. Must be left ON
15-16	Variable Valve	Turns the flowrate of the Carbon Dioxide and Water off/on respectively
17	Air Flow Meter	Measures the total flow of Air going into the system in L/min
18	Carbon Dioxide Flow Meter	Measures the total flow of Carbon Dioxide going into the system in L/min
19	Water Flow Meter	Measures the total flow of water going into the system in L/min
20-21	Go/No-go Valves	Stops/allows flow to system
22	Gas Absorption Column	Packed Column as a part of apparatus
23	Switches	Used to turn on the heater (25), and the water pump (28)
24	Go/No-Go Valves	Allows flow from tank to regulator
25	Heater	Heats CO ₂ to prevents freezing in the system
26	Regulator Valve	Regulates flow of CO ₂
27	CO ₂ Tank	Stores the CO ₂ used to be in the system.
28	Pump	Used to pump water into the column
29	Water Tank	Hold the water, which is pumped out, and recycled back into the tank
30	Regulator Valve	Regulates flow of CO ₂
31	Go/No-Go	Allows/stops flow of air into system

Commented [AZ7]: Rephrase; directs flow from one tube to another...

Commented [AZ8]: Also to extract air from the column

Commented [AZ9]: Yes, use the valve in the back of the unit to adjust air flowrate.

Commented [AZ10]: Water flowrate; the CO₂ flowrate is varied with #18

Commented [AZ11]: #20 is the valve to providing the bottom column pressure to the manometers. #21 is to take a water sample for titration analysis.

Commented [AZ12]: Better name is “cylinder valve”. It is the main valve on a high pressure cylinder.

Commented [AZ13]: Include ‘two-stage pressure regulator’ here.

Commented [AZ14]: More commonly called a ‘high pressure cylinder’

Commented [AZ15]: This is the shutoff valve for the compressed air supply

Commented [PTH16]: Needle valve used to control the flow of air. Between 30 and 31 is the compressed air pressure regulator.

2.2 Materials and Supplies

Table 2.2 Materials and Supplies

Materials/Supplies	Description
NaOH	Used in the Hampel apparatus
50 mL Buret and Buret Clamp	Used during titration
50 mL Graduated Cylinder	Used for determining void fraction of packed bed
Stopwatch	Measures time to reach steady state
125 mL Erlenmeyer flasks	To calculate free CO ₂ in water
Thermometer	For recording temperature of the water
Funnel	For adding NaOH into the system
Gloves	Safety
Lab Glasses	Safety
Stir Bar	Used for titration
Mixer	Used to agitate the stir bar

Commented [PTH17]: 4/5

Commented [PTH18]: also missing CO₂ and compressed air and water

Commented [PTH19]: you're missing parafilm. You need it to cover your sample and prevent CO₂ from escaping

Commented [PTH20]: it's called stir plate but ok

2.3 Experimental Plan

Dependent Variables: CO₂ Concentration, Temperature, Pressure Drop, Mass Transfer Coefficients, and Flooding Point

Independent Variables: Flowrate of CO₂, Flowrate of Water, and Flowrate of Air

Procedure

Commented [PTH21]: 10/10

Part 1: Characterization of Liquid and Gas Flow in Packed Column

Determine the void volume of the packed bed:

- Measure the amount of water needed to immerse a given volume of packing.
OR
 - The second method requires determining the density of solid packing
 - Use the volume to determine the void fraction by having volume of voids/total volume of the packed bed.
 - Density of packing=mass/displacement volume and packed bed density=volume occupied by known weight.
1. Close the drain valve, the control valve, C4, and column drain valve D1.
 2. Fill liquid reservoir tank $\frac{3}{4}$ full of DI water (record temperature)
 3. Adjust air flow in F2 to 20L/min using valve C2.
 4. Dry the column packing by letting air through for 5min
 5. Measure pressure drop across bed for different three flowrates with no liquid
 6. Open C4. C2 and C3 closed. Start liquid pump. Open C1 a couple turns for 12-30s (expels air from pump).
 7. Adjust water flow through column to 1.5 L/min on flowmeter F1 by adjusting C1.
 8. Open C2 and adjust F2 to 20L/min (measure gas pressure drop as air flowrate is increased until flooding)
 9. Repeat for 3L/min, 6L/min and 8L/min

Part 2a: Bringing water to equilibrium with CO₂ and analysis of CO₂ in Water

- 1) Close the control valve (C4) and the column drain valve (D1). Then fill the liquid reservoir tank from the base of the column to three-quarters full of deionized water.
- 2) Record the temperature of the water and place the cover on the tank.
- 3) Open C4 while the gas control valves C2 and C3 are closed and start the liquid pump. Open valve C1 for 15-30 seconds to expel air from pump.
- 4) Adjust the water flow of the column to 6L/min on flowmeter (F1) using C1 valve
- 5) Adjust the air flow rate through F2 to 20L/min using C2. Monitor air flow rate in case of any shifts in flow rates and adjust accordingly.
- 6) Open the main valve for the carbon dioxide cylinder completely. Adjust the pressure by using the regulating valve to about 20psi
- 7) Use valve C3 to give a CO₂ flow rate on F3 to the flow rate assigned. (Between 2 and 5L/min.
- 8) After 10 minutes of operation, take 50mL samples at appropriate intervals from S4 until dissolved CO₂ levels are constant
- 9) Analyze the samples using Analysis Procedure (Appendix B)
- 10) Measure the volume fraction of CO₂ in the air leaving the column from S2 once the dissolved CO₂ levels are constant. (Hempl Apparatus)

Part 3: Absorption of CO₂ by H₂O in continuous column mode

1. Close drain valve, C4, and D1.
2. Fill the tank with DI (Record temperature of water.)
3. Collect sample and determine CO₂ concentration following Analysis Procedure (Appendix B).
4. Open C4. C2 and C3 closed, start the liquid pump. Open C1 couple turns for 15-30sec to expel air. Adjust water flow to 1.5L/min. Open D1. Close c4. Use D1 to maintain liquid water height below the air/CO₂ inlet but above the S4. Leave S4 open.
5. Place bucket under sample tubing to collect water. Adjust air flow rate through F2 to 20L/min using C2.
6. NOTE: monitor and readjust air flow (write them down)
7. Open main valve on CO₂ all the way.
8. Adjust pressure regulating valve on CO₂ regulator of 20psi.
9. Adjust C3 to give CO₂ flow rate on F3

Project Safety Evaluation

Project Title: Gas Absorption in a Packed Bed

Date: January 21st, 2020

	<i>Yes/No</i>	
Potential electrical hazards?	Yes	The pumps when water is in contact may cause a fire of mechanical failures.
Potential mechanical hazards?	Yes	High pressure CO2 gas cylinder, toxic by inhalation and is a simple asphyxiant
Potential pressure hazards?	Yes	High pressure gas exceeding allowed temperature may cause mechanical failure
Potential temperature hazards?	Yes	Water can increase temperature throughout the lifespan of the system running, therefore replacing the water after experiment 1 is required.
Hazardous/toxic chemicals involved?	Yes	NaOH is flammable and may cause eye damage and skin burns.
Gloves required?	Yes	Nitrile gloves should be worn when handling NaOH and titrating
MSDS reviewed by all team members?	Yes	Water [CAS 7732-18-5], Carbon Dioxide [CAS 124-38-9], NaOH [CAS 1310-73-2]
<i>Process Parameters</i>	<i>Max Expected</i>	<i>List location and possible hazards and precautions</i>
Temperature (°C)	25	No heat source in this system
Pressure (psia)	40	CO2 gas pressurized cylinder
<i>Safety Controls</i>	<i>yes/no</i>	<i>If yes, Provide description</i>
Pressure Control	Yes	Pressure regulator and gauges are near the CO2 cylinder.
Level Control	Yes	Water reservoir level should be properly monitored to prevent mechanical failure.

I have read relevant background material for the Unit Operations Laboratory entitled: “Gas Absorption in a Packed Bed” and understand the hazards associated with conducting this experiment. I have planned out my experimental work in accordance to standards and acceptable safety practices and will conduct all of my experimental work in a careful and safe manner. I will also be aware of my surroundings, my group members, and other lab students, and will look out for their safety as well.

Eveyln Almonte
signature of team member

January 21st, 2020
date

Emir Asani
signature of team member

January 21st, 2020
date

Erik Yspango
signature of team member

January 21st, 2020
date

Unit Operations
Group No.2

ChE 382Spring 2020
1/21/2020

Commented [PTH22]: 8/10

Commented [AZ23]: A hazard but not a mechanical hazard.

Commented [PTH24]: mechanical failure of what?

Commented [AZ25]: What other pressure hazards? (tube bursting).

Commented [AZ26]: Put CO2 toxicity hazard here.

Commented [AZ27]: State the max CO2 cylinder pressure (~800 psig) and max compressed air pressure (120psig).

Commented [PTH28]: switch off for the pump needs to be listed out as well.

Commented [AZ29]: Gas shutoff valves, pump and heater switch.

Commented [PTH30]: the heating block for CO2 is also part of the safety controls.

Commented [PTH31]: mechanical failure of what?

Eduardo Salgado
signature of team member

January 21st,2020
date

Micheal Stacy
signature of team member

January 21st,2020
date